Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal
B. 40 Kcal, 5 Kcal
C. 30 Kcal, 8.58 Kcal
D. 25 Kcal, 8 Kcal
E. 29 Kcal, 7.77 Kcal
F. 34 Kcal, 6.9 Kcal
G. 30 Kcal, 6.45 Kcal
H. 32 Kcal, 7 Kcal
I. 35 Kcal, 7.5 Kcal
J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think> Let's solve the problem step by step.

Part (a): Heat required to convert a liter of HâĤĤO at 30Â°C to a liter of water at 60Â°C

1. **Heat required to raise the temperature of 1 liter (1000 g) of water from 30Â°C to 60Â°C:**
- Specific heat capacity of water = 1 cal/gÂ°C
- Mass of water = 1000 g
- Temperature change = 60Â°C - 30Â°C = 30Â°C
- Heat required = mass ÃĹ specific heat ÃĹ temperature change = 1000 g ÃĹ 1 cal/gÂ°C ÃĹ 30Â°C = 30,000 cal = 30 Kcal

2. **Heat required to convert 1 liter of liquid water at 30Â°C to 1 liter of water vapor at 60Â°C:**
- Latent heat of vaporization of water = 540 cal/g (approximately 540 cal/g at 60Â°C, which is a reasonable approximation)
- Heat required = mass ÃĹ latent heat = 1000 g ÃĹ 540 cal/g = 540,000 cal = 540 Kcal

Part (b): Heat required to heat a 1 kg block of aluminum from 30Â°C to 60Â°C

1. **Heat required to raise the temperature of 1 kg (1000 g) of aluminum from 30Â°C to 60Â°C:**
- Specific heat capacity of aluminum = 0.215 cal/gÂ°C
- Mass of aluminum = 1000 g
- Temperature change = 60Â°C - 30Â°C = 30Â°C
- Heat required = mass ÃĹ specific heat ÃĹ temperature change = 1000 g ÃĹ 0.215 cal/gÂ°C ÃĹ 30Â°C = 6450 cal = 6.45 Kcal

Combining the results:
- Total heat required = 30 Kcal + 540 Kcal = 570 Kcal
- Heat required for aluminum = 6.45 Kcal

However, the options provided do not include the total heat for the water conversion and aluminum heating as a single value. Instead, we need to look at the options which combine the results. The closest option is:

<answer> C. 30 Kcal, 8.58 Kcal </answer> </think>